

Shreyansh Sheth

Biomedical Engineering | Neurotechnology | Rehabilitation Engineering

Fredericton, NB, Canada +1 (647) 864-6740 shreyanshsheth5@gmail.com linkedin.com/in/shreyansh-i

RESEARCH PROFILE

Biomedical engineering graduate with advanced training in biomedical engineering and regulatory affairs, seeking a funded research master's as preparation for doctoral study. Research interests centre on neurotechnology, rehabilitation engineering, and biomedical signal processing, with project experience spanning biosignal analysis, assistive systems, and sensor-based diagnostics.

EDUCATION

Master of Science in Regulatory Affairs Northeastern University	2021–2023 GPA 3.28 / 4.00
Master of Technology in Biomedical Engineering Vellore Institute of Technology, Vellore	2020–2022 GPA 3.56 / 4.00
Bachelor of Engineering in Biomedical Engineering Government Engineering College, Gandhinagar Gujarat Technological University	2016–2020 GPA 3.08 / 4.00

RESEARCH INTERESTS

Neurotechnology • Rehabilitation Engineering • Biomedical Signal Processing

SELECTED PROJECTS

Lower-Limb Exoskeleton for Scissoring Gait Developed a theoretical assistive-system design focused on lower-limb rehabilitation and gait support.	Pattern Recognition in ECG Signals through Benford's Law Explored Benford's law as a pattern-recognition approach for electrocardiogram signal analysis.
P300 Signals Review Reviewed P300 event-related potential research and presented the work at ICAICTSEE-2020.	EEG-Based Stress Identification System Used electroencephalography signals and LabVIEW in a system for identifying stress-related patterns.
Smartphone-Sensor Diagnostics Contributed to sensor-based diagnostic research as a Research Assistant at IIIT Raipur.	Automated Medicine Dispenser Built a Raspberry Pi and Python-based system to support scheduled medicine dispensing.
Fingertip Motion Replicating Rehabilitation System Developed a rehabilitation-oriented concept for replicating fingertip motion.	

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Academic & Research Curriculum Vitae

RELATED EXPERIENCE

Team Lead — Regulatory Affairs Co-opApr–Jul 2023

Breegi Scientific Inc. | Boston, MA

- Led three peer students in a four-person final-semester regulatory affairs project.
- Managed project coordination, task planning, and team delivery.
- Supported regulatory work for a Phase I clinical trial involving a neonatal incubator.

Regulatory Affairs Specialist InternJun 2022–Mar 2023

Neuro-Spinal Innovation Inc. | Mississauga, ON

- Supported medical-device risk management activities aligned with ISO 14971.
- Coordinated testing activities with TÜV.
- Contributed to regulatory work involving IEC 60601 electrical-safety testing requirements.

Research AssistantMay–Aug 2021

International Institute of Information Technology, Raipur | Raipur, India

- Contributed to a research project investigating diagnostic uses of smartphone-sensor data.
- Worked with mobile-sensing concepts at the intersection of signal analysis and healthcare diagnostics.
- Supported the project's research activities during a four-month assistantship.

Biomedical Service EngineerJun 2020–Jul 2021

Sterling Hospitals | Ahmedabad, India

- Supported service activities for biomedical equipment in a hospital environment.
- Applied biomedical engineering training in an equipment-focused clinical setting.
- Contributed to the hospital's biomedical equipment support function.

TECHNICAL SKILLS

Analysis	MATLAB, R
Programming	Python
Engineering Tools	LabVIEW, COMSOL
Regulatory Affairs	Medical-device regulatory submissions; risk management per ISO 14971; IEC 60601 testing
Focus Areas	Biomedical signal processing, EEG, ECG, sensor-based diagnostics, assistive systems

SELECTED CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- g.tec BCI & Neurotechnology Spring School, 2021
 - TCPS 2: Course on Research Ethics
 - NIH Good Clinical Practice (GCP)
- Human Anatomy Specialization — University of Michigan
 - MATLAB Masterclass
 - COMSOL Training

CONFERENCE PRESENTATIONS

Analyzing P300 SignalsICAICTSEE-2020

Stress Identification SystemISC-2021

LANGUAGES

English: Fluent | Hindi: Fluent

Gujarati: Fluent | French: Intermediate (B1)